Chapter test

2 Molecules of life

Name: _____ ( ) Class: _____ Date: _____

Total marks: 40 marks
Time allowed: 40 minutes
Answer ALL questions.

Score: / 40

A. Concept checking (10 marks, 1 mark each)

1. What are the three most common elements in living organisms?

   A. oxygen, hydrogen and nitrogen
   B. nitrogen, carbon and phosphorus
   C. oxygen, hydrogen and carbon
   D. carbon, sulphur and nitrogen ☐

2. Water is important to humans because

   (1) it provides turgidity to maintain the body shape.
   (2) it acts as a medium of transport in the human body.
   (3) it helps maintain a relatively constant body temperature.

   A. (1) and (2) only
   B. (1) and (3) only
   C. (2) and (3) only
   D. (1), (2) and (3) ☐

3. Which of the following descriptions about minerals is ***incorrect***?

   A. They are essential to the regulation of metabolism.
   B. They are broken down during respiration to provide energy to organisms.
   C. They are inorganic in nature.
   D. They are transported by water in the human body. ☐

4. Which of the following statements about glycogen is correct?

   A. It is a storage form of carbohydrate in animals.
   B. It is a component of plant cell walls.
   C. It can form brick-red precipitate in Benedict's test.
   D. It is a polysaccharide found in both animals and plants. ☐

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★ 5. Which of the following substances is the most soluble in water?

A. sucrose
B. cellulose
C. glucose
D. glycogen ☐

6. What is the function of cholesterol in the human body?

A. as a component of cell membranes
B. protecting internal organs
C. as an insulating layer to reduce heat loss
D. as an energy reserve ☐

7. Which of the following chemical substances can provide energy to the body?

(1) carbohydrates
(2) lipids
(3) minerals

A. (1) and (2) only
B. (2) and (3) only
C. (1) and (3) only
D. (1), (2) and (3) ☐

8. The specificity of amino acids is determined by the

A. amino group.
B. side chain.
C. carboxyl group.
D. peptide bond formed between amino acids. ☐

★ 9. Which of the following is ***not*** the primary function of proteins?

A. as cell component
B. growth and tissue repair
C. form enzymes
D. energy storage ☐

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10. Which of the following comparisons about DNA and RNA is correct?

	DNA	*RNA*
A.	contains ribose	contains deoxyribose
B.	single stranded	double stranded
C.	contains uracil	contains thymine
D.	relatively stable	relatively unstable

☐

Question	1	2	3	4	5	6	7	8	9	10
Related section	2.1	2.2	2.3	2.4	2.4	2.5	2.5	2.6	2.6	2.7

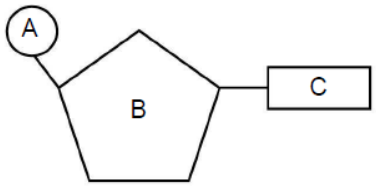
B. Problem solving (22 marks)

11. For each of the minerals listed in column 1, select the corresponding function(s) from column 2. Put the appropriate letter(s) in the space provided.

Column 1		*Column 2*
(a) Calcium	_____	A. For muscle contraction
(b) Magnesium	_____	B. Forming haemoglobin
(c) Iron	_____	C. Forming chlorophyll
		D. Component of bones and teeth

(Total: 3 marks)

12. The diagram below shows a basic unit that makes up deoxyribonucleic acids.



(a) Identify the components labelled A, B and C. (3 marks)

_____

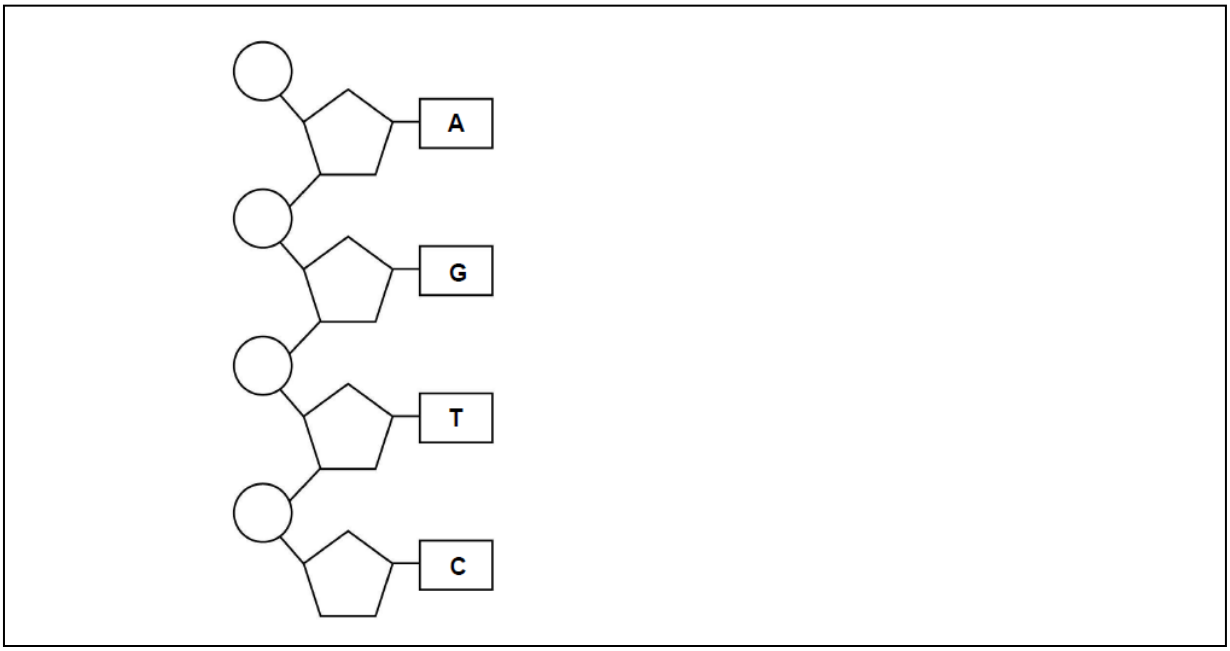
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(b) Describe the structure of a DNA molecule. (3 marks)

(c) In a DNA molecule, component C pairs up in specific combinations: adenine (A) pairs with thymine (T), and guanine (G) pairs with cytosine (C). The diagram below shows one of the polynucleotide chain of a DNA molecule. Draw the complementary polynucleotide chain to complete the diagram. (2 marks)



(d) Give ***two*** differences between DNA and RNA. (2 marks)

(e) Give ***one*** function of DNA and RNA, respectively, in organisms. (2 marks)

(Total: 12 marks)

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★ 13. John wants to build muscles. He bought an energy bar from the supermarket and carried out several food tests on it. The results are summarized in the table below.

Food tests	Results
Glucose strip test	No colour change
Benedict's test	Precipitate formed
Iodine test	Mixture turned blue-black
Grease spot test	Translucent spot appeared
Protein strip test	No colour change

(a) What kinds of carbohydrates are found in the energy bar? (2 marks)

(b) The appearance of a translucent spot is not enough to determine the presence of lipids in the energy bar. Based on the property of lipids, describe and explain what further steps should be carried out to confirm the presence of lipids in the energy bar. (3 marks)

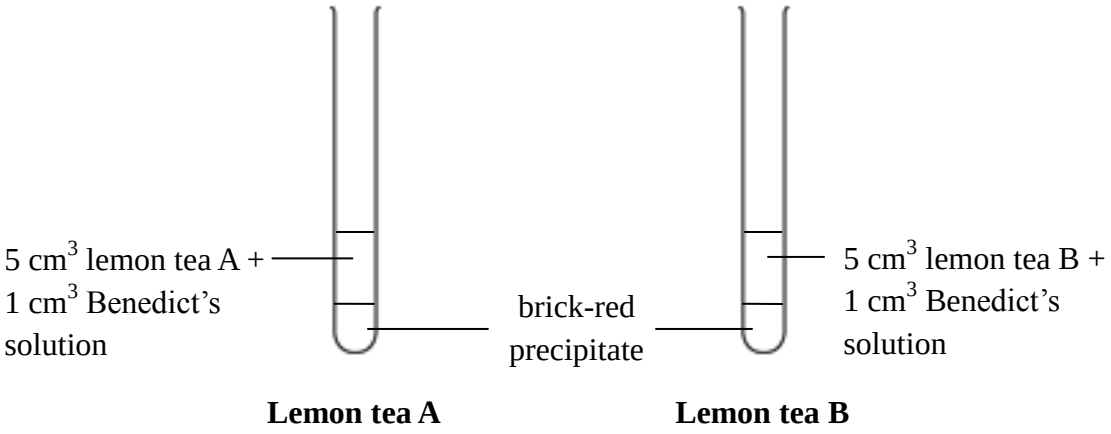
(c) Can this energy bar help John to build muscles? Why? (2 marks)

(Total: 7 marks)

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C. Exam challenge (8 marks)

★★ 14. A student carried out Benedict's test to detect the amount of reducing sugar in two brands of lemon tea, A and B. The results are shown below:



(a) Describe how the student can compare the amount of precipitate formed when the difference in the height of precipitate in both set-ups is not obvious. (4 marks)

(b) The food labels of the two brands of lemon tea are shown below.

LEMON TEA A	
Nutrition Facts	
Serving 1 carton (250 mL)	
Calories	150
Fat	0 g
Saturated fat	0 g
Carbohydrates	40 g
Sugars	40 g

Lemon Tea B	
Nutrition Facts	
Serving	1 carton (250 mL)
Calories	185
Fat	0 g
Saturated fat	0 g
Carbohydrates	60 g
Sugars	60 g

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(i) Suggest why the experimental results were different to the nutrition facts shown above. (2 marks)

(ii) The student suggested that consuming large amount of these brands of lemon tea would not cause overweight since they do not contain any fat. Do you agree with him? Explain your answer. (2 marks)

(Total: 8 marks)

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Marking scheme

A. Concept checking (10 marks, 1 mark each)

1.	C	2.	C	3.	B	4.	A	5.	C
6.	A	7.	A	8.	B	9.	D	10.	D

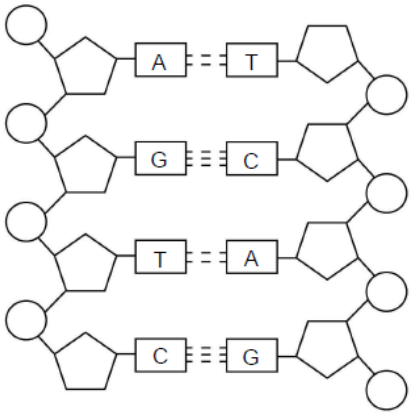
(1 mark each)

B. Problem solving (22 marks)

11. (a) A, D (1)
    (b) C, D (1)
    (c) B, C (1)

(Total: 3 marks)

12. (a) A is the phosphate group. (1)
    B is a deoxyribose. (1)
    C is a nitrogenous base. (1)
    (b) A DNA molecule consists of two polynucleotide chains (1)
    coiled around each other to form a double helix. (1)
    The two chains are held together by hydrogen bonds. (1)
    (c) Correct base pairing (1)
    Correct polynucleotide chain (1)



    (d) The 5C sugar in DNA is deoxyribose while that in RNA is ribose. (1)
    A DNA molecule consists of two polynucleotide chains while an RNA molecule consists of a single polynucleotide chain only. (1)
    (e) DNA stores genetic information. (1)
    RNA is involved in protein synthesis. (1)

(Total: 12 marks)

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13. (a) reducing sugars (except glucose) (1)
starch (1)
(b) Add drops of organic solvent on the translucent spot. (1)
Lipids are insoluble in water but soluble in organic solvents. (1)
If the translucent spot contains lipids, the spot would disappear after drying. (1)
(c) No. This energy bar does not contain any proteins, (1)
which are needed for building muscles. (1)

(Total: 7 marks)

C. Exam challenge (8 marks)

14. (a) Filter the mixture with a filter paper. Rinse the test tube with distilled water a few times and filter the rinsing in order to obtain all the precipitate. (1)
Dry the precipitate together with the filter paper in the oven. (1)
Measure the original weight and the final weight of the precipitate together with the filter paper. Calculate the weight of the precipitate. (1)
Repeat the steps with the other sample and compare the weight difference of the precipitate formed. (1)
(b) (i) The amount of Benedict's solution was not enough to react with all reducing sugar. / Reducing sugar in the lemon tea was in excess. (1)
Not all reducing sugars were oxidized. (1)
(ii) No. These brands of lemon tea contain high sugar content, (1)
Excess sugar will be converted to fat and stored in the body, causing overweight. (1)

(Total: 8 marks)

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